COPD, IN ONE BREATH

WHAT IS COPD?

A common, preventable and treatable disease with persistent respiratory symptoms and airflow limitation. Breathlessness, cough and sputum are typical. It is not curable, but it is treatable.

WHAT IS AN EXACERBATION?

An acute worsening of respiratory symptoms that leads to additional therapy — from extra bronchodilators at home up to hospitalisation. Exacerbations drive health status, admissions and disease progression.

WHY IS LONGITUDINAL MONITORING USEFUL?

A single value is a moment. Repeated records against a personal baseline begin to show direction of change — and the strongest known predictor of future exacerbations is the history of past ones.

WHEN YOU ANSWER

Answer in one breath.

Then stop talking.

Do not add a number you cannot defend.

If you do not know, say: “We have not evaluated that — it would be part of the next stage.”

SOURCES

WHO COPD fact sheet, 10 June 2026; GOLD 2026 report.
GOLD 2026 report.
Hurst JR et al., N Engl J Med 2010;363:1128-38.

RESEARCH PROTOTYPE · SYNTHETIC DATA ONLY NOT A MEDICAL DEVICE · NOT CLINICALLY VALIDATED

THE DIGITAL TWIN

WHAT IS THE DIGITAL TWIN IN PRANAQUARD?

A versioned, replayable state per patient: observations, a personal baseline from the first 72 observed hours, derived features, and the model context attached to each saved state. It is data-driven, not a mechanistic lung model.

WHAT DOES THE MODEL ACTUALLY PREDICT?

An experimental estimate of a new synthetic exacerbation onset within 24, 48 or 72 hours on the research cohort. On held-out synthetic data the 72-hour model reached AUROC 0.395 and AUPRC 0.059 — below chance and below prevalence. Reported as weak and non-clinical.

WHAT DATA IS REAL AND WHAT IS SYNTHETIC?

Every patient, physiological stream and outcome label is synthetic — 120 generated patients. The only real inputs are public modelled city air-quality and weather data, used as context, not personal exposure. No clinical data is involved, so nothing here is clinical validation.

WHEN YOU ANSWER

Answer in one breath.

Then stop talking.

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If you do not know, say: “We have not evaluated that — it would be part of the next stage.”

SOURCES

PranaGuard docs/METHODOLOGY.md, docs/ARCHITECTURE.md. artifacts/evaluation.md; docs/MODEL_CARD.md. docs/DATA_PROVENANCE.md; docs/LIMITATIONS.md.

EVIDENCE AND HONESTY

HAS THE SYSTEM BEEN CLINICALLY VALIDATED?

No. No clinical trial, no real patient data, no regulatory review. It is labelled a research prototype and not a medical device throughout the product.

WHAT HEALTHCARE IMPACT HAS ACTUALLY BEEN DEMONSTRATED?

Software impact only: a working, audited, reproducible prototype with tested clinician and patient workflows. No patient, clinical or system-level outcome has been measured.

HOW WOULD THE PLATFORM BE CLINICALLY EVALUATED?

Three stages: clinician-supervised usability and workflow studies; prospective observational research with adjudicated outcomes under ethics approval; then outcome measurement including calibration, subgroup fairness and net benefit.

WHEN YOU ANSWER

Answer in one breath.

Then stop talking.

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If you do not know, say: “We have not evaluated that — it would be part of the next stage.”

SOURCES

docs/LIMITATIONS.md; README caution block.
DELIVERY_REPORT.md; docs/LIMITATIONS.md.
docs/LIMITATIONS.md (next steps).

THE CLINICIAN WORKFLOW

HOW DOES THE DOCTOR INTERACT WITH THE PLATFORM?

Signs in, opens an assigned patient, reviews the twin, the trends and the labelled model context, then records a review. Access is enforced server-side by patient assignment, and every action is attributed in an audit trail.

WHEN YOU ANSWER

Answer in one breath.

Then stop talking.

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If you do not know, say: “We have not evaluated that — it would be part of the next stage.”

SOURCES

docs/DOCTOR_PATIENT_DELIVERY.md.

IF THE ROOM PUSHES BACK

IS THIS JUST A DASHBOARD?

No. A dashboard shows the latest number. This keeps every hour as a versioned, replayable state, so the clinician can look at direction of change against that patient's own baseline.

WHY SHOULD WE TRUST A MODEL THAT SCORES 0.4?

You shouldn't trust it as a clinical tool — and we don't ask you to. The value we demonstrated is that the workflow is buildable and auditable. The model is a research component with published, honest limitations.

SO WHAT WOULD YOU NEED TO MAKE THIS REAL?

Ethics approval and prospective data with adjudicated outcomes, then calibration and subgroup analysis, then outcome measurement. That is a multi-year research programme, not a software release.

WHEN YOU ANSWER

Answer in one breath.

Then stop talking.

Do not add a number you cannot defend.

If you do not know, say: “We have not evaluated that — it would be part of the next stage.”

SOURCES

All answers traceable to project documentation and cited publications.